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Studierichting: Informatica
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$$f(x) = \frac{1}{x^2 + 2x}$$

$$f'(x) = -\frac{2x + 2}{(x^2 + 2x)^2}$$

$$f''(x) = -1 \cdot \left[\frac{2(x^2 + 2x)^2 - (2x + 2)2(x^2 + 2x)(2x + 2)}{(x^2 + 2x)^4}\right]$$

$$= -1 \cdot \left[\frac{2(x^2 + 2x)((x^2 + 2x) - (2x + 2)^2)}{(x^2 + 2x)^4}\right] = -1 \cdot \left[\frac{2(x^2 + 2x - 4x^2 - 8x - 4)^2)}{(x^2 + 2x)^3}\right]$$

$$= \frac{2(3x^2 + 6x + 4)}{(x^2 + 2x)^3}$$

$$f'''(x) = 2\left[\frac{(x^2 + 2x)^3(6x + 6) - (3x^2 + 6x + 4)3(x^2 + 2x)^2(2x + 2)}{(x^2 + 2x)^6}\right]$$

$$= 2\left[\frac{(6x + 6)(x^2 + 2x)^2((x^2 + 2x) - (3x^2 + 6x + 4))}{(x^2 + 2x)^6}\right]$$

$$= 2\left[\frac{(6x + 6)((x^2 + 2x) - (3x^2 + 6x + 4))}{(x^2 + 2x)^4}\right]$$

$$= 2\left[\frac{(6x + 6)(-2x^2 - 4x - 4)}{(x^2 + 2x)^4}\right] = -24\left[\frac{(x + 1)(x^2 + 2x + 2)}{(x^2 + 2x)^4}\right] = \frac{-24(x^3 + 3x^2 + 4x + 2)}{(x^2 + 2x)^4}$$

References